OBJECT DETECTION BASED ON FASTER R – CNN USING RANGE COMPRESSED AIRBORNE RADAR DATA

PROBLEM STATEMENT

Identifying a ship's lifespan using radar data is challenging, but feasible with range-compressed airborne radar data and convolutional neural networks. Collaboration between experts, data scientists, and engineers is crucial to develop robust methodologies for ship lifespan identification in maritime surveillance.

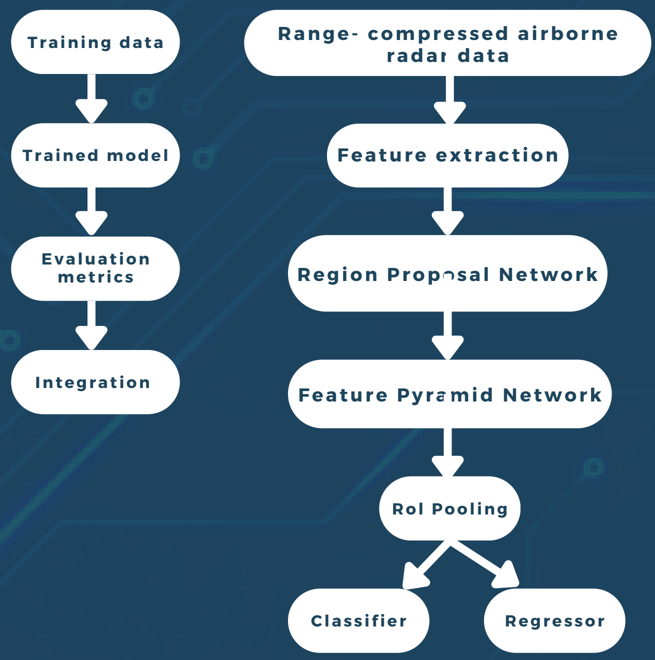
OBJECTIVE

Developing two ship detection algorithms using Faster RCNN's for airborne radar data analysis, one for time and one for the Doppler domain. Training on a comprehensive dataset for robustness, with a focus on near real-time capabilities, CFAR challenge mitigation, and integration with AIS and marine radars.

APPLICATIONS

1. Interoperability
2. Interaction
3. Heterogeneity
4. Scalability
5. Performance Optimization
6. Security and Privacy
7. Deployment and Monitoring
8. Maintainability and Extensibility

SYSTEM ARCHITECTURE



RESULTS

Near real-time ship monitoring is crucial. Airborne radars and Faster RCNN detectors show promise. Future focus: multi-sensor integration, real-time fusion, machine learning, and cybersecurity for advanced maritime safety.

REFERENCES

1. Revised Guidelines for The Onboard Operational Use of Shipborne AIS, Int. Maritime Org., London, U.K., 2015, pp. 1-17.
2. A. Moreira, P. Prats-Iraola, M. Younis, G. Krieger, I. Hajnsek, and K. P. Papathanassiou, "A tutorial on synthetic aperture radar," IEEE Geosci. Remote Sens. Mag., vol. 1, no. 1, pp. 6-43, Mar 2013.